# silent shepherd.

ears for your machines

# Hearing the Unheard

From the whisper of **wind** predicting a **storm** to the subtle **sound** of a broken **twig**, our **ears** are ancient **sentinels**, often our very last line of **defense** against an incoming **danger**.

Long before **sight** confirms, **sound** alerts – a primal **alarm system** finely tuned to work even in our **sleep**. This innate ability to listen, interpret, and respond has been **fundamental** to human **survival**.

Yet, the **machines** we build, despite their incredible capabilities, are often **deaf** to their own plight.

In each case, the **failure** begins with an **unheard warning**.

# The Cost of Failure: A 3D Printing Nightmare

In a **large scale** industrial 3D printing facility, a **critical** aerospace part runs on a **14-hour** print cycle at **95%** capacity.

Hours before completion, a **nozzle** begins to **clog**. The warning appears first as a **subtle** change in **sound** and **micro-vibration**—but no **operator** or **sensor** detects it.

The print **fails** catastrophically, wasting both **material** and **14 hours of work**, forcing **unscheduled downtime**, and **missing critical deliveries**. One failure cascades into **lost revenue**, **production delays**, and **damaged reputation**.

# silent shepherd.

We provide a **solution** that can intelligently **analyze** a 3D printer's unhealthy **behaviors** and react **immediately**.

Using **AI powered** audio analysis and **real-time** audio **monitoring**:

Most **failures** are **heard** before they are **seen**

- clogs, wear, slip, and misfeeds change sound first.

**Cameras** and **sensors** react too **late**

- after geometry or output is already compromised.

Acoustic monitoring is **passive, non-intrusive**, and **scalable**

- enabling early detection across entire facilities.

Making **audio** a powerful and **underused** sensing **layer** across physical infrastructure.

# product.

Real-time acoustic monitoring detects 3D print failures before they destroy the part.

- Microphone array with beam steering
- AI audio analysis (real-time)
- Integrated camera
- Instant alerts

# values.

**Catch** failures at hour 2, not hour 14

**Save** $500-2000 per **prevented failure**

Planned **maintenance**, not emergency **shutdowns**

One **operator** monitors **10+ printers**

# market.

The numbers tell a stark story about why Silent Shepherd matters.

## 20.8B

### 3D Printing Market

The industrial 3D printing market is growing **~15% annually** and expected to reach **USD 20.8 B in 2026**.

## 125K+

### Downtime Cost

Unplanned manufacturing downtime typically costs **~$125,000–$260,000 per hour** in lost production value.

## 100K+

### Single Failure Impact

A single failed long print (e.g., nozzle clog) can cascade into **$100K–$1M+ in losses** through wasted hours, delays, and backlog impact, eroding revenue and customer trust.

# competition.

## Squarehead Technologies (Norway)

- Hardware-only, expensive upfront

## 3DSignals (Israel)

- Proprietary sensors required, high cost

## Augury (USA)

- Vibration-only, doesn't work in noisy environments

## Arundo Analytics (USA)

- Generic approach, not 3D printing optimized

## silent shepherd. (Latvia)

- AI audio cleaning
- Beam steering
- Integrated camera
- 3D printing optimized
- Lower cost

# business model.

## SaaS

- $50-100/month per printer
- Recurring revenue

## Hardware

- Microphone arrays: $200-400
- Camera modules: $150-250
- One-time revenue

## unit economics

### CAC

~$2000

### LTV

$15,000+

### Payback

2-3 months

### Gross Margin

75%+ SaaS

# go-to-market & validation.

## Strategy

1. Target mid-to-large 3D printing shops (50+ printers)
2. 3-month free pilot on 5 printers
3. Prove ROI, then expand to full fleet
4. Aggregate data, improve models, build moat
5. Expand to CNC, injection molding, ship engines

## Proof

- 6-week pilot: detected 12 failures, saved $150K+ in wasted material
- 15+ shop owners interviewed: 100% confirmed nozzle clogs are #1 pain
- All expressed willingness to pay $50-100/month per printer

# financial timeline.

## YEAR 1 (PRE-SEED)

- Raise €200K pre-seed
- Goal: create a functioning, testable prototype
- Activities: ongoing real-world testing, interviews, securing local pilot partnerships
- Focus: validate product-market fit and prove technical functionality
- Revenue: minimal (early pilot testing only)

## YEAR 2 (SEED ROUND)

- Raise seed round
- Expand team for deep product development
- Goal: complete pilot integrations with early partners
- Start generating first real recurring revenue
- Reach 20 recurring clients

## YEAR 3–5 (GROWTH PHASE)

- Full product launch
- Drive recurring revenue through subscription growth (priced at €20/mic/month)
- Optional mid-range hardware sales (€100 per mic)
- Emphasize scalable architecture and minimal human deployment
- Head towards 1000 recurring clients

# team.

## Edvards — CEO / COO

- 10+ years leading complex audio projects across film, TV, and installations
- Founder of BRODERS AUDIO, managing production teams and clients
- National award-winner for sound design and project execution

## Teodors — CTO

- Built audio AI and embedded systems at Neural DSP and MikroTik
- Invented and developed industry-recognized Light Pedal product
- Led academic R&D in spatial audio and neural audio modeling

## Krisjanis — COO / CEO

- Worked at Merito Partners, supporting private equity deal analysis
- Led financial planning and investor materials at SAGA Space Architects
- Top student in SSE Riga’s Financial Economics course

# the ask.

## €200,000 Pre-Seed Round

We are raising €200,000 to finance 12 months of operations. This round will fund full-time work by the three co-founders and allow for aggressive prototyping of both hardware and software components. The goal is to validate product-market fit and prepare for scalable deployment.

## milestones

- Launch trials with 5 pilot clients
- Demonstrate real-time anomaly detection
- Build foundation for scalable product rollout

## next round

Next round planned in 12 months, based on product traction and trial results.

# let's give ears to our machines.

We invite visionary investors to join us in shaping the future of industrial reliability. Connect with our founders to learn more about this groundbreaking opportunity.


### Edvards — CEO

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